

STANDARD OPERATING PROCEDURE

HOW TO MANAGE BACKEND DATA UPDATE REQUEST

REVISION HISTORY

Revision No./Version No.	Date	Prepared by	Approved by	Details of Change
1.0	27-Oct-2023	Prachi Naik (Asst. Manager IT)	Anant Sham Yende (CTO)	Initial Release
2.0	01-July-2025	Prachi Naik (Asst. Manager IT)	Anant Sham Yende (CTO)	Addition of Workflow
3.0	27-Jan-2026	Prachi Naik (Asst. Manager Technical)	Anant Sham Yende (CTO)	Addition of check for key field and non-key field. Addition of Review & approval Matrix

OBJECTIVE

To ensure all the update request received to update any type of data in backend given either by clients, applicants or PM are received with proper update information & necessary documentation. To also ensure such update requests are kept for reference whenever needed.

PARTICIPANTS

1. Project Manager (PM): PM on behalf of client or data owner can request for updating.
2. Implementation Engineer (IE): IE on behalf of Project Manager(PM)/Junior PM/Assistant Executive can request for updating.(provided this request is reviewed by PM)
3. TL/ Engineer: engineer who will update as per the request made.

INTRODUCTION

I. Understanding back-end data update requests.

1. What is back-end data update request?

It's a request raised to the software development/maintenance team (specifically the database team) to update the data maintained in the system directly from back-end without using the front-end interface to accommodate the requirements of the client.

2. Which data can be considered for Back-end Update Request?

The data update can be considered into below two categories,

1. Data received in bulk for addition/update/deletion E.g.

- a. Update disbursement amount for all the Griha Aadhar beneficiaries from Rs.1200/- to Rs.1500/-
- b. Update status of NOC for water connection applications as On Hold for the list of ack nos attached
- c. Update application records between X & Y dates as disposed.

2. Individual record to be updated

- a. Update the boat type from Parasailing Boat to Passenger Boat for application acknowledgement no. ACK272730

3. What are key/critical field & Non-Key fields?

Key/Critical fields are data elements that directly impact core functionality, business logic, financial outcomes, compliance requirements, or overall service delivery.

Examples include: bank account number, beneficiary name, benefit amount, benefit category, disbursement status, payment mode, and master data used for financial transactions.

Non-Key fields are data elements that do not directly impact the core functionality, financial outcomes, compliance, or critical business logic of a service. Updates to these fields have limited or no impact on service delivery and do not affect disbursement, calculations, or regulatory requirements.

4. Why a backend data update needed or what is the REASON for backend update?

- 1. The software interface doesn't allow user to update data from front-end interface due to
 - a. Field is disabled or non-editable
 - b. It's a system generated field and cannot be changed
 - c. No provision in the front-end interface to delete a record.

2.The applicant/end user has made a mistake while applying/processing information that cannot be changed once submitted.

E.g. the system doesn't allow to update the name of the applicant after submission. The officer has approved the certificate and wants to rollback same to correct some content in the certificate.

3. To change the state of the record to a new state, than what is available in the system as per client requirement. (Update data to override the system's basic flow and make the records available for further processing for department). e.g If the application is pending with certain verification officer and due to their absence the director puts a request to mark the application to other verification officer.

4. Data is more and needs to be added/updated from the spreadsheet/flat files/google sheets and there is no bulk data management provision available in the system in the front end.

5.The system has wrongly processed data due to bug or coding error. (NOTE: For this additionally GEL Jira ticket needs to be raised under production issue.)

5. Who can request back-end data update?

1.The data update can be requested by client/data owner through a letter/email

2.The Project Manager can also request on behalf of the client.

6. Who can authorize back-end data update requests?

1. Non-Key Field Update (Applicable to both Bulk Data Update and Individual Record Update):
 - a. Client authorisation, Project Manager (PM) authorisation, and Team Lead (TL) authorisation are mandatory for any updates to non-key fields (whether bulk or individual).
 - b. All authorisation must be documented and attached to the update request for audit and traceability purposes.
 - c. No update shall be performed without the complete set of authorisation from all relevant parties.
2. Key / Critical Fields Update (Applicable to both Bulk Data Update and Individual Record Update):
 - a. A formally documented note detailing the reason for change, impact analysis, along with client approval, Project Manager (PM) approval, and Team Lead (TL) approval is mandatory.
 - b. CEO approval is compulsory and must be documented and attached to the respective ticket.
 - c. No update to any key or critical field shall be performed without explicit CEO approval.

II. Pre-requisites

1. A Written request from the client/Applicant/Project Manager (PM)/Junior PM/Assistant Executive
2. The request should mention clearly the following details:
 - a. Why does data need to be updated or which reason category does the update belong to (refer Introduction section point no 3)
 - b. Which data needs to be updated & when data update needs to be carried out in the system
 - c. Whether the update in any way impact financial benefits (sanction/disbursement/recovery etc)

PROCEDURE TO FOLLOW

1. PM should raise a GELJIRA ticket towards "Back-end data Update request" issue type. PM should attach a client request letter and any supporting act/rules and also mention clearly why, which, when towards the data to be updated if not already mentioned in the client letter.
2. The implementation engineer can as well raise a "Backend data update request" however, the same needs to be confirmed by respective project manager by putting their comments on the ticket and marking to TL.
3. The ticket raised should clearly mention whether updation is towards Key field or Non-Key field.
4. TL should first check the request as below:
 1. Check if the update is towards a key field or non key field.
 2. Check for client approval/request document and approval of PM in form of comment is present.
 3. Check the GELJira ticket for clear understanding of what data is to be updated, when it should be done and why the data needs to be updated.
 4. Go through the attached documents and check if any discrepancies in request from department letter and the ticket raised. If so, bring it to the notice of the project manager.
 5. If the data to be added/updated is received in spreadsheet/flat file do following checks
 - a. Check if proper column names are mentioned or any ambiguity.
 - b. Check if data is corrupted or lost formatting or shifted columns in spreadsheet
 - c. Check if numbers are being shown as exponential
 - d. Check if columns have unrelated values
 - e. If multiple sheets with similar data, merge sheets by verifying the column order and add an extra column at last to mention the reference of the records pulled from different sheets.
 - f. Whether update is applicable for multiple records in the database
 - g. Whether update is related to financial benefits/sanctions/disbursement
 6. In case of any ambiguity or improper information/request, revert immediately to the Project manager.
 7. **NO ASSUMPTIONS** to be done on the data or requested updates. If the request doesn't state clearly what to update and where to update, and when to update don't proceed. Mark the ticket back to PM for clarification with proper questions.
 8. If updation is towards **non key fields** then proceed and allot to developer.

9. If Updation is towards **key fields** prepare the impact and identify dependencies & issue and revert the same on ticket to PM. PM to now get the approval of CTO & CEO via Note approval process.
 10. In case of **non key fields** if all approvals are valid TL to proceed and in case of **key fields** TL to proceed only when note approval is uploaded on ticket.
 11. Once clear on what to be done, allot a developer to do the task.
 12. Cross check the query written by the developer and validate the records being updated before committing
 13. Always audit current data being updated
 14. Add the reference to the data being updated by mentioning the ticket in the update reason.
 15. Always select data to be updated before updating to check if proper and same number of records are being updated.
 16. Always use transaction locks to control or rollback to avoid any data loss.
 17. Below details need to be shared in case of bulk and individual requests respectively to the PM before doing any updation
- I. For Bulk update**
 - a. How many records will be updated
 - b. List of the records that will be updated
 - c. Any impact that will be caused due to the update.
 - II. For Individual update**
 - a. Snapshot of the record that will be updated.
18. Upon approval from PM, carry of the requested updation.
 19. After activity validate if the required changes are affected properly
 20. After updating the spreadsheet request data, generate the updated data in the same format as requested and export to Excel and counter check using VLOOKUP or compare with update requested original file.
 21. The TL needs to share the outcome of the update after carrying out the update in the database for the understanding of the PM/Support team and for onward submission to client for their reference.

Below details need to be shared in case of bulk and individual requests respectively

III. For Bulk update

- a. How many records updated
- b. List of the records updated drawn from the database
- c. Any impact due to the update.

IV. For Individual update

- a. Updated record snapshot
22. The PM needs to communicate the status of the update request with the client along with the details of updated data as mentioned above.

TIMELINE TO FOLLOW

Basic updates to be processed in 24 hours after following due procedures

Category	Update Complexity	Low Priority [Hrs]	Medium Priority [Hrs]	High Priority [Hrs]
(bulk)	Low	24	16	8
	Medium	32	24	16
	High	48	32	24
(individual)	Low	8	4	2
	Medium	16	8	4
	High	24	16	8

NOTE: In case, update request is of high priority, the Project Manager should have a call scheduled with the technical lead and brief on the GELJira ticket and clarify doubts if any.

Backend Data Update Review and Approval Matrix

Type	Client	TL	PM	CTO	CEO
Non-Key Fields	Authorization to update data via mail/letter/CR	1.Review & proceed only if client authorization and PM approval are attached. 2.Review request, assess impact & highlight dependencies/issues that will occur due to update and seek PM approval on Ticket	Review & approve update by commenting on ticket.	—	—
Key Fields	Authorization to update data via mail/letter/CR	1.Review & proceed only if client authorization and PM approval are attached. 2.Review request, assess impact & highlight dependencies/issues that will occur due to update and submit for PM approval.	Review request and takes impact from TL and marks note to CTO for approval.	Confirms the update & then forward to CEO	Final approval

Logging in Jira/TM

Issue Type*: Backend Data Update Request

Assignee*: TL

Reporter (ticket to be created by) *: PM/IE

Priority*:High/Medium/Low

Description: Fill in the details requested as per below

1. Whether Non-Key field or Key Field.
2. Updation details:
3. Any specific conditions to update (mention if any) :

Work Logging by*:TL/Developer

Note: A Backend Data Update Request ticket should be **closed by the IE/ticket creator** only after confirming that the update has been completed and notifying the client or the requester who initiated the request.

Gel-Jira Ticket Workflow for Backend Data Update Request

